

DNA / PROTEIN SEQUENCE ANALYSIS USING BLAST REPORT

Title: Homology Analysis of Human KLK3 (Prostate-Specific Antigen) Protein Using BLASTP

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1. Introduction

Bioinformatics utilizes computational tools to analyze biological sequences and investigate evolutionary relationships among organisms. One of the most widely used tools for sequence comparison is the Basic Local Alignment Search Tool (BLAST), which identifies homologous sequences in biological databases.

The protein selected for this analysis was **KLK3 (Kallikrein Related Peptidase 3)**, commonly known as **Prostate-Specific Antigen (PSA)**. KLK3 is a serine protease produced primarily by prostate epithelial cells and is widely used as a biomarker in prostate cancer diagnosis and monitoring.

The objective of this study was to identify homologous protein sequences related to human KLK3 and evaluate their similarity using BLASTP.

2. Materials and Methods

Sequence Retrieval

The KLK3 protein sequence was obtained from the National Center for Biotechnology Information (NCBI).

Gene Name: KLK3

Gene ID: 354

Protein Accession: NP_001639.1

Protein Length: 261 amino acids

BLAST Analysis

The protein sequence was submitted to the NCBI BLASTP tool using the default parameters.

Program: BLASTP

Database: ClusteredNR

The resulting homologous sequences were evaluated based on:

- Percent Identity
- Alignment Score
- Query Coverage
- E-value

3. Results

Conserved Domain Analysis

BLAST identified a conserved domain belonging to the:

Tryp_SPc (Trypsin-like Serine Protease Domain)

This finding is consistent with the biological function of KLK3 as a serine protease enzyme.

Top Homologous Sequences

Species	Protein	Identity (%)	Positives (%)	E-value
<i>Homo sapiens</i>	Prostate-specific antigen isoform 1	100	100	0.0
<i>Pan paniscus</i>	Prostate-specific antigen	99	99	4×10^{-179}
<i>Pongo pygmaeus</i>	Prostate-specific antigen isoform X	97	98	0.0
<i>Papio anubis</i>	Prostate-specific antigen precursor	90	95	1×10^{-173}
<i>Macaca fascicular</i>	Prostate-specific antigen	89	95	6×10^{-169}

Alignment Statistics

The query sequence showed extremely high similarity to prostate-specific antigen proteins from several primate species.

The human KLK3 sequence demonstrated:

- Score: 536 bits
- E-value: 0.0
- Identity: 261/261 (100%)
- Positives: 261/261 (100%)
- Gaps: 0%

These values indicate a perfect alignment with the reference sequence.

Figure 1: DNA/Protein Sequence Analysis BLAST Output

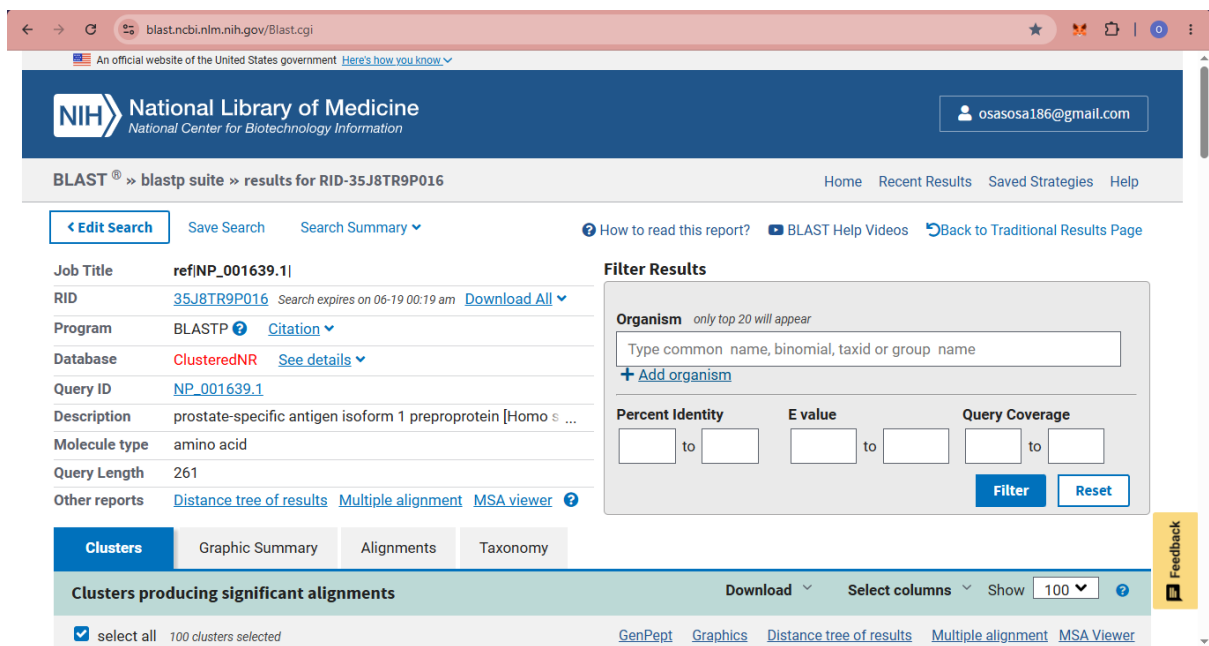


Figure 2: DNA/Protein Sequence Analysis Graphic Summary

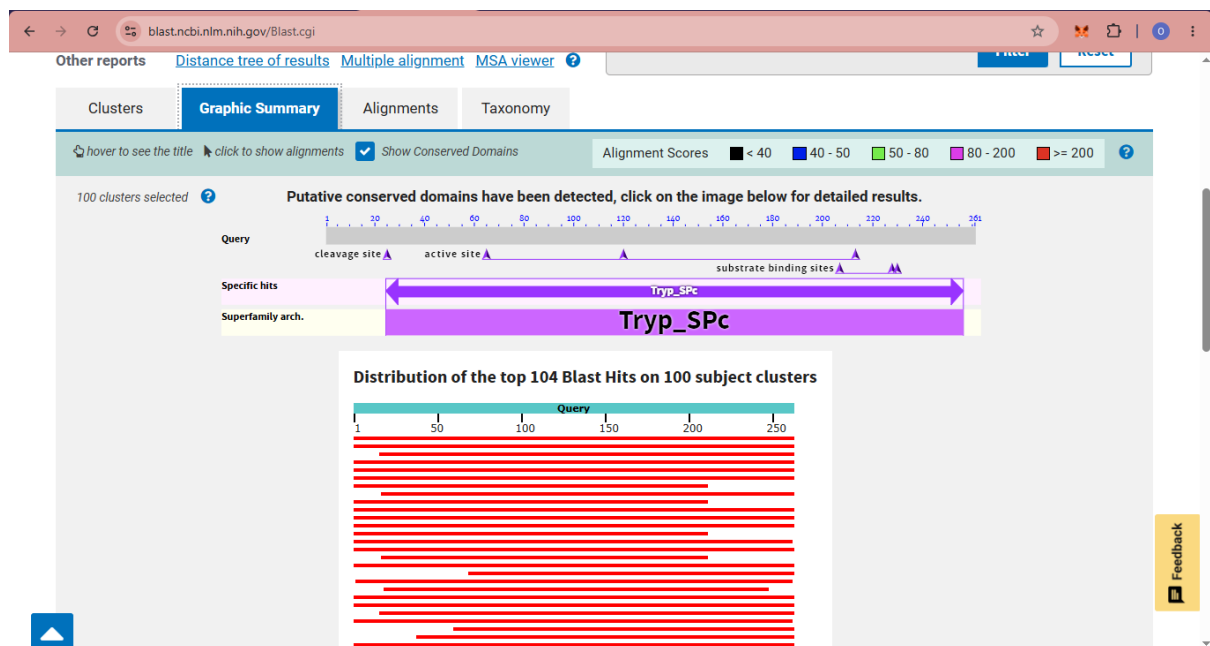
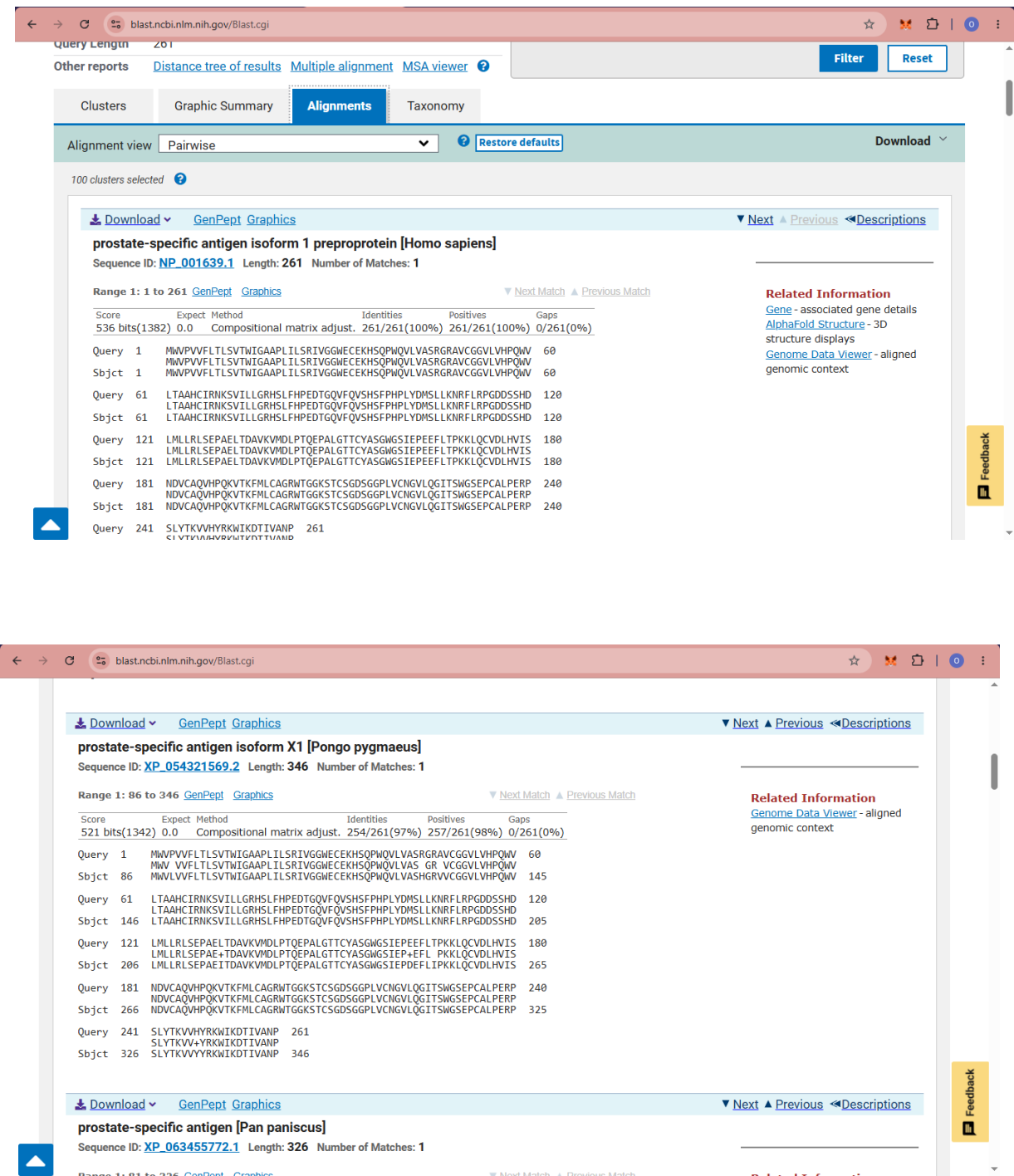


Figure 3: DNA/Protein Sequence Analysis Alignments



blast.ncbi.nlm.nih.gov/Blast.cgi

Sbjct 326 SLYTKVYVYRKWKIDTIVANP 346

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prostate-specific antigen [Pan paniscus]

Sequence ID: [XP_063455772.1](#) Length: 326 Number of Matches: 1

Range 1: 81 to 326 GenPept Graphics

Score	Expect	Method	Identities	Positives	Gaps
505 bits(1301)	4e-179	Compositional matrix adjust.	243/246(99%)	244/246(99%)	0/246(0%)

Query 16 GAAPLILSRIVGGWECEKHSQPWQVLVASRGRAVCGGVLVHPQWVLTAAHCIRNKSIVILL 75

Sbjct 81 GAVPLILSRIVGGWECEKHSQPWQVLVASRGRAVCGGVLVHPQWVLTAAHCIRNKSIVILL 140

Query 76 GRHSLFHPEDTGQVFQVSHSFPHPLYDMSLLKNRFLRPGDDSSHDMLRLRSEPAELTDA 135

Sbjct 141 GRHSLFHPEDTGQVFQVSHSFPHPLYDMSLLKNRFLRPGDDSSHDMLRLRSEPAELTDA 200

Query 136 VKVMDLPTQEPALGTTTCYASGWSIEPEEFLTPKKLQCVDLHVISNDVCAQVHPQKVTKF 195

Sbjct 201 VKVMDLPTQEPALGTTTCYASGWSIEPEEFLTPKKLQCVDLHVISNDVCAQVHPQKVTKF 260

Query 196 MLCAGRWITGGKSTCSGDSGGPLVCNGVLQGITSWGSEPCALPERPSLYTKVHYRKNWIKD 255

Sbjct 261 MLCAGRWITGGKSTCSGDSGGPLVCNGVLQGITSWGSEPCALPERPSLYTKVHYRKNWIKD 320

Query 256 TIVANP 261

Sbjct 321 TIVANP 326

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prostate-specific antigen precursor [Papio anubis]

Sequence ID: [NP_001106216.1](#) Length: 261 Number of Matches: 1

Feedback

Figure 4: DNA/Protein Sequence Analysis Clusters

blast.ncbi.nlm.nih.gov/Blast.cgi

Clusters Graphic Summary Alignments Taxonomy

Clusters producing significant alignments

Download Select columns Show 100

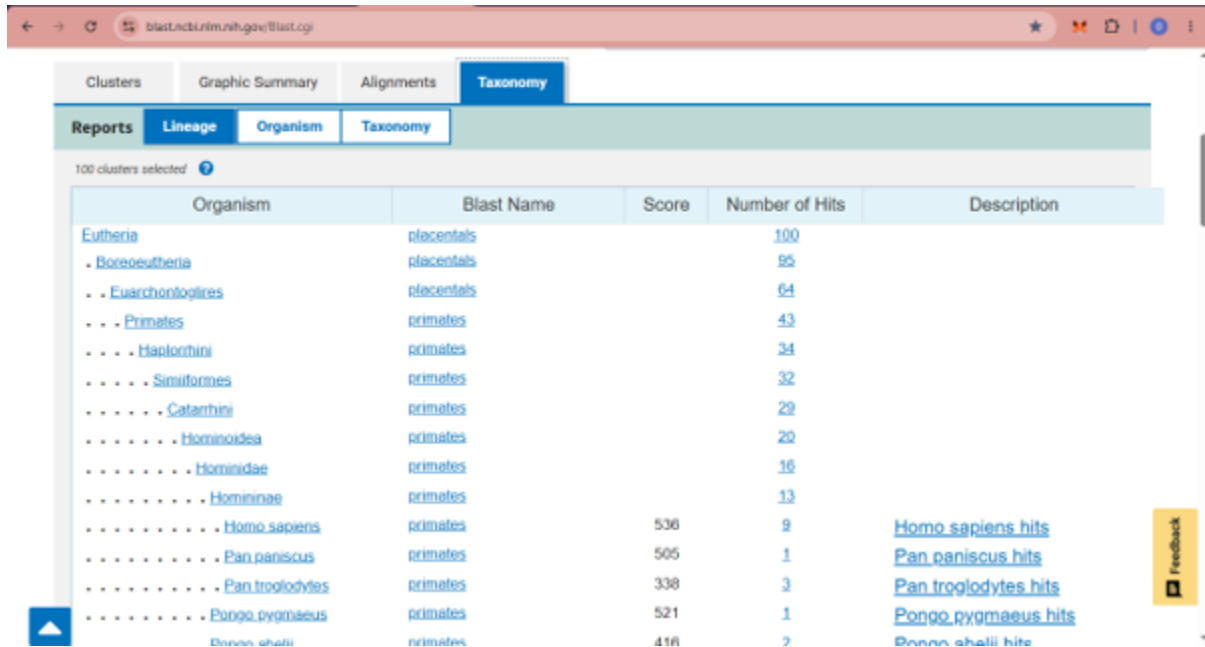
☒ select all 100 clusters selected

GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

	Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	33 member(s), 21 organism(s)	primates	prostate-specific antigen isoform 1 preproprotein [Homo sapiens]	536	536	100%	0.0	100.00%	261	NP_001639.1
☒	2 member(s), 2 organism(s)	great apes	prostate-specific antigen isoform X1 [Pongo pygmaeus]	521	521	100%	0.0	97.32%	346	XP_054321569.2
☒	1 member(s), 1 organism(s)	pygmy chimpanzee	prostate-specific antigen [Pan paniscus]	505	505	94%	4e-179	98.78%	326	XP_063455772.1
☒	13 member(s), 12 organism(s)	Old World monkeys	prostate-specific antigen precursor [Papio anubis]	489	489	100%	1e-173	90.42%	261	NP_001106216.1
☒	6 member(s), 5 organism(s)	Old World monkeys	prostate-specific antigen [Macaca fascicularis]	481	481	100%	6e-169	89.27%	375	XP_005590104.4
☒	1 member(s), 1 organism(s)	silvery gibbon	prostate-specific antigen isoform X3 [Hylobates moloch]	452	452	100%	2e-159	84.27%	248	XP_058282011.1
☒	6 member(s), 5 organism(s)	apes	kallikrein 3 (prostate specific antigen) isoform CRA_b [Homo sapiens]	432	432	80%	6e-152	100.00%	213	EAW71923.1
☒	2 member(s), 1 organism(s)	siamang	prostate-specific antigen isoform X2 [Symphalangus syndactylus]	432	432	94%	5e-151	86.53%	285	XP_063471005.1
☒	12 member(s), 11 organism(s)	primates	prostate-specific antigen isoform 3 preproprotein [Homo sapiens]	431	431	80%	5e-151	100.00%	238	NP_001025218.1
☒	5 member(s), 5 organism(s)	apes	prostate-specific antigen isoform 4 preproprotein [Homo sapiens]	425	425	100%	4e-149	83.14%	218	NP_001025219.1
☒	21 member(s), 14 organism(s)	primates	kallikrein-2 isoform 1 preproprotein [Homo sapiens]	419	419	100%	8e-146	77.01%	261	NP_005542.1
☒	2 member(s), 2 organism(s)	primates	kallikrein-2 isoform X1 [Pongo abelii]	416	416	100%	1e-143	76.63%	354	XP_024094143.3
☒	1 member(s), 1 organism(s)	Rhesus monkey	prostate-specific antigen isoform X2 [Macaca mulatta]	390	390	80%	4e-135	90.00%	238	XP_014979905.3
☒	1 member(s), 1 organism(s)	Bolivian squirrel monkey	kallikrein-2 [Saimiri boliviensis]	397	397	100%	3e-134	73.46%	475	XP_003940458.2
☒	2 member(s), 2 organism(s)	primates	PREDICTED: kallikrein-2-like isoform X2 [Propithecus coquerelli]	385	385	100%	8e-133	68.97%	261	XP_012498601.1

Feedback

Figure 5: DNA/Protein Sequence Analysis Taxonomy



100 clusters selected

Organism	Blast Name	Score	Number of Hits	Description
Eutheria	placentals		100	
• Boreoeutheria	placentals		95	
• • Euarchontoglires	placentals		64	
• • • Primates	primates		43	
• • • • Haplorhini	primates		34	
• • • • • Simiiformes	primates		32	
• • • • • • Catarrhini	primates		29	
• • • • • • • Hominoidea	primates		20	
• • • • • • • • Hominidae	primates		16	
• • • • • • • • • Homininae	primates		13	
• • • • • • • • • • Homo sapiens	primates	536	9	Homo sapiens hits
• • • • • • • • • • Pan paniscus	primates	505	1	Pan paniscus hits
• • • • • • • • • • Pan troglodytes	primates	338	3	Pan troglodytes hits
• • • • • • • • • • Pongo pygmaeus	primates	521	1	Pongo pygmaeus hits
• • • • • • • • • • • Oranorhina	primates	416	2	Oranorhina hits

4. Discussion

The BLAST analysis revealed that KLK3 is highly conserved among primates. The highest similarity was observed between human KLK3 and corresponding prostate-specific antigen proteins in bonobo and orangutan, exhibiting identities of 99% and 97%, respectively.

The very low E-values indicate that these alignments are statistically significant and unlikely to have occurred by chance. Such strong conservation suggests that KLK3 performs important biological functions that have been maintained throughout primate evolution.

The conserved Tryp_SPc domain identified during the analysis further confirms the classification of KLK3 as a serine protease. Conservation of this catalytic domain across species indicates preservation of enzymatic function.

These findings demonstrate how BLAST can be used to identify homologous proteins and investigate evolutionary relationships among organisms.

5. Conclusion

BLASTP analysis of the human KLK3 protein successfully identified multiple homologous sequences among primate species. High sequence identities and significant alignment scores indicate strong evolutionary conservation of prostate-specific antigen proteins.

The presence of the conserved Tryp_SPc serine protease domain further supports the functional importance of KLK3 across species. This exercise demonstrates the usefulness of bioinformatics tools in sequence analysis, protein characterization, and comparative genomics.